Lesson 3: Presentation Design Principles

Most technical presentations fail in the first two slides. The author knows the material cold, the data is solid, and the meeting is scheduled with the right stakeholders — and then the deck opens with a wall of text on a cluttered title slide, followed by a diagram copied from a wiki that nobody can read at the back of the room, and the audience has already checked out. The material was never the problem. The deck was. This lesson is about the design discipline that separates a deck that communicates from a deck that simply contains information.

You will stay inside the **OrbitCo** scenario from Module 1. The 30-day ticket review you built in Lesson 1, and the one-page dashboard you built in Lesson 2, are about to become a briefing deck. The dashboard's Mean Time To Resolve (MTTR) tile becomes a slide element. The weekly trend chart becomes an embedded, annotated visual. The Service Level Agreement (SLA) breach view becomes the evidence block behind a recommendation. Lesson 4 takes the deck you structure here and turns it into a five-minute briefing — the capstone of this course. Use of productivity tools (Excel, PowerPoint) for technical reporting is the Registered Training Institute (RTI) topic in play across both lessons of this module, and this one focuses on the PowerPoint half.

Two competencies from the A-28 On-the-Job Learning (OJL) standard are tagged against this lesson. The first is **2d — use and maintain runbooks and incident response documentation** — because the technical-diagram skills you practice here (network topology, flow diagrams, escalation paths) are the same skills that produce the visuals inside a real runbook. The second is **5d — develop and maintain technical documentation, Standard Operating Procedures (SOPs), and runbooks** — because a well-designed PowerPoint deck with a master layout and consistent theming is itself a form of living technical documentation when it is reused for status meetings, onboarding, or quarterly reviews. Both competencies develop on the job; what this lesson gives you is the artifact-creation skill that makes the documentation worth reading when the time comes.

Learning Objectives

By the end of this lesson, you will be able to:

1. Structure a technical presentation using the problem → evidence → options → recommendation pattern
2. Apply slide design best practices (layout, hierarchy, typography, contrast) to improve readability
3. Use PowerPoint layouts, master slides, and themes to enforce consistency across a deck
4. Create technical diagrams (network topology, flow diagrams, architecture blocks) using shapes and SmartArt
5. Embed Excel charts and tables into slides without losing fidelity or linking incorrectly

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Structuring a Technical Presentation

A well-structured deck does not ask the audience to follow the author's analytical path. It presents the findings in the order a decision-maker needs them, which is often the reverse of the order in which the author discovered them. Structure is the first and most important design decision, and most decks lose the audience because structure was skipped in favor of starting to make slides.

The Problem → Evidence → Options → Recommendation Pattern

This four-part pattern is the default structure for any technical presentation that ends in a request — a hire, a budget, an architectural change, a staffing shift, a policy update. It works because it maps to how humans evaluate proposals. First you establish that there is a problem worth solving. Then you show the evidence that supports it. Then you lay out the options that were considered. Then you recommend the one you believe is right and explain why.

For the OrbitCo 30-day review, the structure writes itself:

- **Problem:** Incident volume is up 22% over the prior period, and MTTR is trending the wrong way on High-severity tickets.
- **Evidence:** The weekly trend chart and the severity-by-team pivot from the Lesson 2 dashboard. The SLA breach count. The uptime number, which is still inside SLA but slipping.
- **Options:** Add one on-call engineer. Refactor the top-three recurring runbooks. Tighten the paging policy. Do nothing and re-evaluate in 30 days.
- **Recommendation:** Refactor the top-three runbooks first (highest-leverage, lowest-cost), then re-evaluate the on-call ask in 30 days with fresh data.

Why Executives Want the Recommendation First

A senior decision-maker often has five to ten minutes for your entire meeting. If you spend the first eight minutes setting up the problem, you may not get to the recommendation at all. The inverted structure — lead with the recommendation, then show the evidence that justifies it — is the executive summary pattern. For this audience, use it. For a technical peer review (where the audience needs to understand

your analysis before they can evaluate your conclusion), the forward structure (problem first, recommendation at the end) is right. The underlying content is identical; only the lead changes.

Standard Sections

A defensible long-form technical deck has these sections, in this order: **title**, **context** (one slide), **findings** (multiple slides, one main finding per slide), **analysis** (the interpretive layer — what do the findings mean?), **recommendation**, **next steps**, **Q&A**. An appendix of supporting material sits after Q&A and is referenced but not presented.

Slide Count Discipline

One main idea per slide. No walls of text. If a slide contains three main ideas, it should be three slides — each with a single clear headline. The most common sign of a poorly-structured deck is a slide whose title is a vague noun ("Analysis," "Results," "Data") instead of a sentence that names the finding.

The 10-Minute Rule

If the material takes more than ten minutes to present, you probably need a handout in addition to slides. Slides are a real-time visual aid; handouts carry detail the audience will read asynchronously. The OrbitCo capstone briefing is five minutes, which means five to seven slides — the slide count is a function of the time budget, not the amount of material.

Slide Design Best Practices

Once structure is settled, design is about making each slide readable in five seconds. The reader's first pass is a glance — if that glance does not yield the main idea, you have lost attention. The techniques below are what make the glance pay off.

Visual Hierarchy

Visual hierarchy is the design principle that the most important element on a slide should be the most visually dominant. A slide has exactly one dominant element — the headline finding, the key chart, the recommendation sentence — and everything else is sized, colored, and positioned to support rather than compete with it.

Signs of broken hierarchy: three charts of equal size on one slide. Two bold headlines. A footer that is visually louder than the body. A logo that dominates the title. If the viewer's eye does not land in one obvious place, the hierarchy has failed.

Typography

The readable defaults are short and sans-serif. Use Calibri, Segoe UI, or Arial for body text. Reserve a second typeface (often a display serif like Georgia) for the deck's title slide only, if at all. Size discipline:

- Body text: 18pt minimum. If you need more room, make a second slide — do not shrink the type.
- Headlines: 28pt minimum. A sentence-case headline at 28pt reads faster than a title-case phrase at 40pt.
- Speaker notes: can be any size you like, since the audience does not see them.

Avoid all-caps body text (it reduces comprehension), italics on long blocks (hard to read on projectors), and more than two typefaces in the same deck.

Color and Contrast

Every foreground and background combination should meet **Web Content Accessibility Guidelines (WCAG) AA contrast** — roughly a 4.5:1 contrast ratio for body text and 3:1 for large text and graphics. Pale grey text on a white background fails this and is invisible on a dim projector.

Avoid red and green as the only distinction between two categories. Roughly 8% of men and 0.5% of women have red-green color vision deficiency; those viewers see your red/green chart as grey/grey. Pair color with another visual cue (shape, label, pattern).

White Space as a Design Element

White space is not empty space. It is the breathing room that lets the eye find the dominant element. A slide that is crammed edge-to-edge forces the viewer to scan and parse before they can interpret. A slide with generous margins and one centered chart produces the interpretation in a second.

Alignment and the Grid

Consistent left-alignment is the default for body content. Centered text is slower to read for anything longer than a headline. Align shape edges to each other, not to the slide's edges — a row of three boxes should share the same top and bottom coordinates exactly. PowerPoint's alignment tools (**Arrange → Align → Align Top / Distribute Horizontally**) do this in two clicks.

Activity: Slide Rebuild

In this hands-on lab, you receive a deliberately bad OrbitCo post-mortem slide — eleven bullets, three fonts, an unreadable screenshot, and a title that reads "Results" — and rebuild it into one or two clean slides that follow the design principles above. The goal is to move from a slide that contains information to a slide that communicates a single clear takeaway at a glance.

Using Layouts, Master Slides, and Themes

PowerPoint's most powerful design features are the ones that enforce consistency across a deck without requiring the author to remember every style choice on every slide. When a deck drifts — three fonts, five title positions, inconsistent bullet styles — the problem is almost never a lack of care. It is that the deck was built without using the master.

The Slide Master

The **Slide Master** (View → Slide Master) is the template behind every slide in the deck. It defines the default fonts, colors, logo placement, footer content, background, and bullet styles. Changes to the master propagate to every slide that uses the master's layouts.

A deck with a well-designed master needs only content edits going forward. The title font, the body color, the footer text — all set once, all consistent. A deck without a master needs style edits on every slide the author touches, which is the main way decks drift over time.

Built-in Layouts

PowerPoint ships with standard layouts: **Title**, **Title + Content**, **Two Content**, **Comparison**, **Title Only**, **Blank**, and a few more. Each is a pre-designed arrangement of placeholders. Use them. Typing directly onto a blank slide instead of choosing a layout is how decks end up with inconsistent title positions.

For the OrbitCo briefing, the layout choices map to the deck structure:

- Title slide → **Title** layout
- Context slide ("Here is where we are as of Day 30") → **Title + Content**
- Weekly trend and severity mix on one slide → **Two Content**
- Options comparison → **Comparison**
- Recommendation → **Title Only** with a single large sentence

Custom Layouts

If a recurring slide pattern does not exist as a built-in layout — for example, a "metrics dashboard" layout with four Key Performance Indicator (KPI) tiles in a 2×2 grid and a chart strip across the bottom — build it once as a custom layout in the Slide Master. Every slide using that layout will be identical. This is the trick behind recurring status decks that stay visually consistent across months of use.

Themes

A **theme** is a coordinated set of fonts, colors, and effects applied across the whole deck. Most organizations have an approved theme; use it. If you are building a deck from scratch, choose one of PowerPoint's built-in themes and stick with it. Mixing theme elements from two different themes is the fastest way to produce a visually incoherent deck.

Inheriting a Messy Deck: The Reset Button

When you are handed a deck that has drifted — slides with off-master fonts, misaligned titles, inconsistent footers — select the affected slides and use **Home → Reset**. Reset snaps each slide back to its layout's master-defined positions, typefaces, and colors without touching the content. It is the single most useful command for cleaning up an inherited deck.

Creating Technical Diagrams

Diagrams are where technical presentations earn their keep. A well-drawn network topology communicates system structure in one image that would take three paragraphs to describe. A clean incident-escalation flow answers "what happens when X fails?" faster than any prose runbook. Every operations engineer produces diagrams as part of their documentation work; getting them right is the OJL 2d point exactly — the runbook visuals you produce in the field depend on the diagram skill you build here.

Shapes, Connectors, and Grouping

The fundamental diagramming primitives in PowerPoint are **shapes** (rectangles, circles, arrows, cloud shapes for the internet boundary) and **connectors** (lines that snap to shape edges and move with the shapes). Use connectors, not free-drawn lines — connectors follow the shape when it moves, which keeps the diagram coherent under edits.

Grouping (Ctrl+G / Cmd+G) combines several shapes into a single group object so they move and resize together. For a three-tier diagram, group the web-tier shapes into one group, the app tier into another, and the data tier into a third — now layout changes are a single drag per tier, not a per-shape drag.

Snap-to-Grid and Alignment

PowerPoint's alignment tools are the difference between a diagram that looks professional and one that looks hand-drawn by accident. **Arrange → Align** with **Distribute Horizontally** and **Align Top** applied to a row of boxes gives them pixel-perfect spacing and alignment. **View → Gridlines** turns on a visible grid you can snap to.

For any diagram with more than three shapes, running Align and Distribute at the end is non-negotiable. It takes ten seconds and catches every "this box is off by two pixels" flaw that would otherwise make the diagram feel sloppy.

SmartArt for Flows, Hierarchies, and Escalation Paths

SmartArt (Insert → SmartArt) is PowerPoint's generator for common diagram templates — hierarchy trees, process flows, Venn diagrams, pyramids, and relationship maps. The power is speed: you type a bullet list, pick a SmartArt style, and PowerPoint produces a formatted diagram.

For the OrbitCo deck, SmartArt is the right tool for the incident escalation path (L1 support → L2 on-call → L3 platform team → Engineering Director), the runbook flow (alert fires → triage → contain → resolve → post-mortem), and the team structure hierarchy tree for the support organization. Each of these is a conventional shape that SmartArt templates cover cleanly.

SmartArt vs Hand-Built Shapes

SmartArt is fast but rigid. You pick a template and your diagram fits the template. Hand-built shapes are slower but let you draw exactly what the system looks like — a network topology with specific nodes, or a three-tier architecture with the real service names.

The rule: SmartArt for generic flows (processes, hierarchies, relationships where the shape is conventional). Hand-built shapes for specific technical architectures where precision matters. An incident escalation flow is SmartArt. A network diagram of OrbitCo's production environment is hand-built.

Consistency in Diagram Notation

Use the same symbol for the same kind of thing everywhere in the deck. Rectangles for services. Cylinders for databases. Cloud shapes for external dependencies. Hexagons or colored highlights for critical components. If an API gateway is a blue rectangle on one slide, it is a blue rectangle on every slide. This is what separates a designed deck from a collection of one-off diagrams.

Labels and Flow Direction

Every node in a diagram gets a label. Connectors get labels wherever the direction or relationship is not obvious from the drawing. Flow direction should be consistent — left-to-right or top-to-bottom for sequential flows, center-outward for hierarchy. A

diagram that mixes directions is harder to read than one that picks a direction and commits.

Activity: Build an OrbitCo System Diagram

In this hands-on lab, you take a written escalation scenario (an OrbitCo P1 ticket flowing through L1 support, L2 on-call, L3 Platform, the Engineering Director, and the post-mortem owner) and produce two versions of the diagram — one using SmartArt, one hand-built from shapes and connectors. You then compare the two and commit to the one you would include in the Lesson 4 briefing.

Embedding Excel Content in Slides

The final design skill for this lesson is getting Excel content — charts, tables, KPI tiles — into PowerPoint without losing fidelity or introducing maintenance debt. The OrbitCo dashboard from Lesson 2 has several visuals that will need to appear in the Lesson 4 briefing; this section is how to move them across.

Copy as Picture: Static and Safe

Right-click the Excel chart and choose **Copy as Picture**, then paste into PowerPoint. The result is a static image that preserves the exact rendering of the chart. You cannot edit the underlying data from inside PowerPoint, which is the point — the image cannot drift from the data because it is decoupled entirely.

Use picture paste for final decks, decks distributed to people who do not own the source, and decks where the data is frozen at a specific point in time (for example, a post-mortem for a specific incident).

Paste as Embedded Excel Object

Paste Special → Microsoft Excel Worksheet Object embeds an editable copy of the source Excel file inside the PowerPoint. The receiver can double-click the chart and edit the underlying worksheet in place.

The cost: file size grows (sometimes dramatically), and the embedded copy has no link back to the source — so edits you make to the original Excel do not propagate to the slide, and vice versa. Use sparingly. Most of the time, either picture paste or linked paste is a better choice.

Paste with Link

Paste Special → Paste Link creates a slide chart that is linked to the source Excel file. When the source workbook updates and the PowerPoint is reopened, the chart reflects the new data.

Linked paste is the right choice for recurring decks — the weekly operations review, the monthly status readout — where the same slide shape gets refreshed data every cycle. It is the wrong choice for any deck that will travel outside the author's

filesystem, because the link breaks the moment the source Excel is not accessible at the original path.

When Links Break

Links break when the source Excel file is renamed, moved, or shared with someone who does not have access. A broken link typically shows up as a red "X" or as a stale version of the chart that will not update. **File → Info → Edit Links to Files** lets you see and repair link paths.

Two patterns keep links healthy: store the source Excel and the PowerPoint in the same folder and reference with relative paths, or keep both in a shared location (OneDrive, SharePoint, a shared team drive) that every intended viewer can reach.

Paste Special → Picture for Font-Portability

If the chart uses a font that may not exist on the target machine, Paste Special → Picture (Portable Network Graphics or Enhanced Metafile) renders the chart using the current machine's fonts at paste time and locks in the visual. The resulting image is not editable but is font-safe.

Wrap-Up

Structure first, then design, then diagrams, then embedded data. A deck built in that order has a clear argument, slides that read at a glance, diagrams that communicate system structure, and data visuals that carry their meaning honestly. The skills in this lesson are the foundation for the five-minute OrbitCo capstone briefing, where the focus shifts from designing the deck to delivering it.

Key Takeaways

- Structure is the first design decision — problem → evidence → options → recommendation for forward presentations; invert the lead for executive audiences but keep the same four components.
- One main idea per slide, with a sentence-style title that names the finding. Walls of text and vague noun titles ("Analysis," "Results") are structural failures, not stylistic ones.
- Use the Slide Master and built-in layouts to enforce consistency. A drifted deck is almost always a deck whose author skipped the master; the Reset button recovers most of the damage.
- Pick SmartArt for generic flows (escalation, process, hierarchy) and hand-built shapes for specific architectures (network topology, named services). Label every connector; align every row.
- Choose paste mode based on where the deck is going: picture for portable one-time decks, linked for recurring shared-folder decks, embedded only when recipients genuinely need to edit the data.

Reflection Questions

1. Think about the last technical slide deck you sat through. Where did its structure break — no problem framing, no clear recommendation, or too many ideas on one slide? What would you redesign first?
2. If you inherited a 20-slide deck with inconsistent fonts, three different title positions, and no Slide Master, what is the order of operations you would follow to clean it up in under 30 minutes?

3. A colleague asks you to embed "the latest" OrbitCo dashboard chart into a quarterly board briefing that will be forwarded outside the company. Which paste mode do you recommend, and what is the one-sentence justification you give them?